Country Integration Service: Architecture and Flow Diagrams

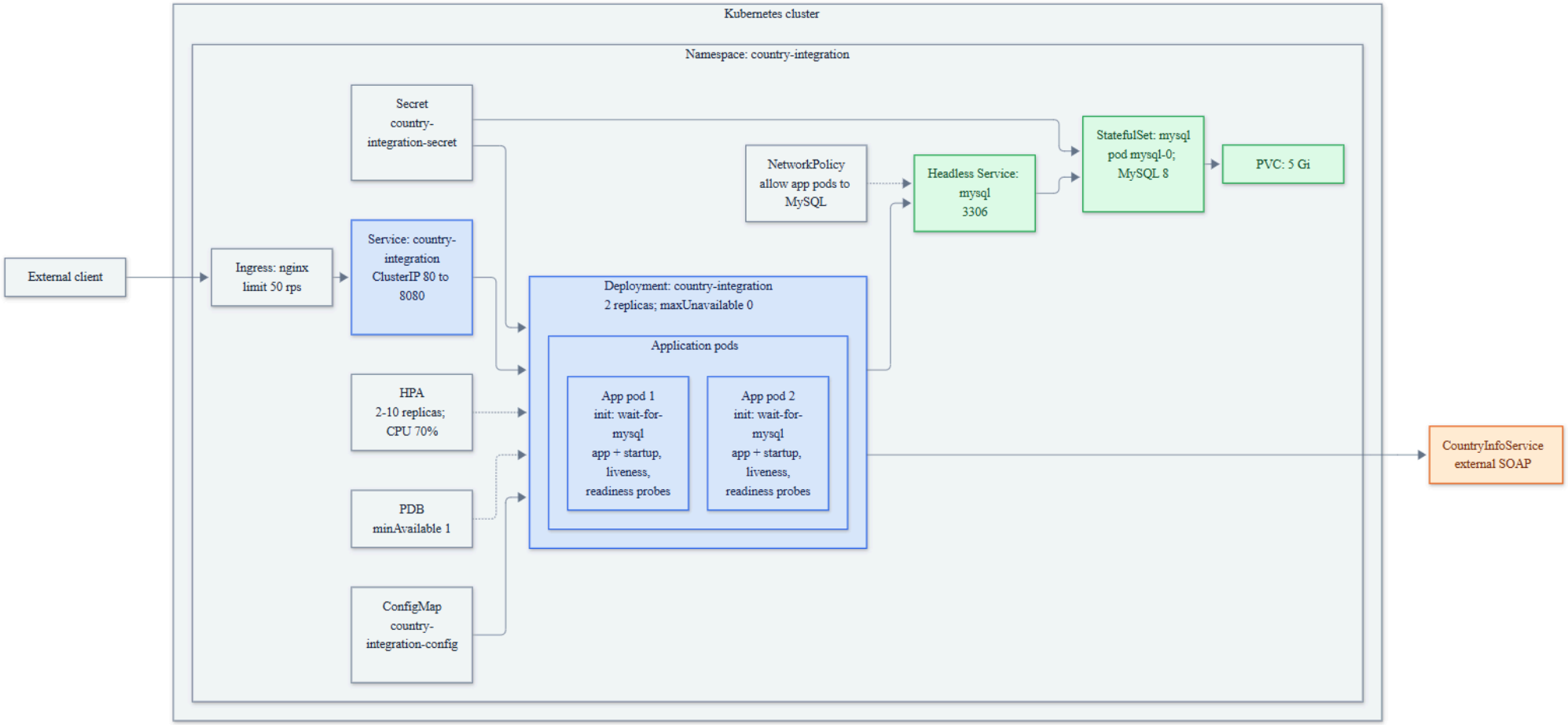
NCBA LOOP case study

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8 October 2026

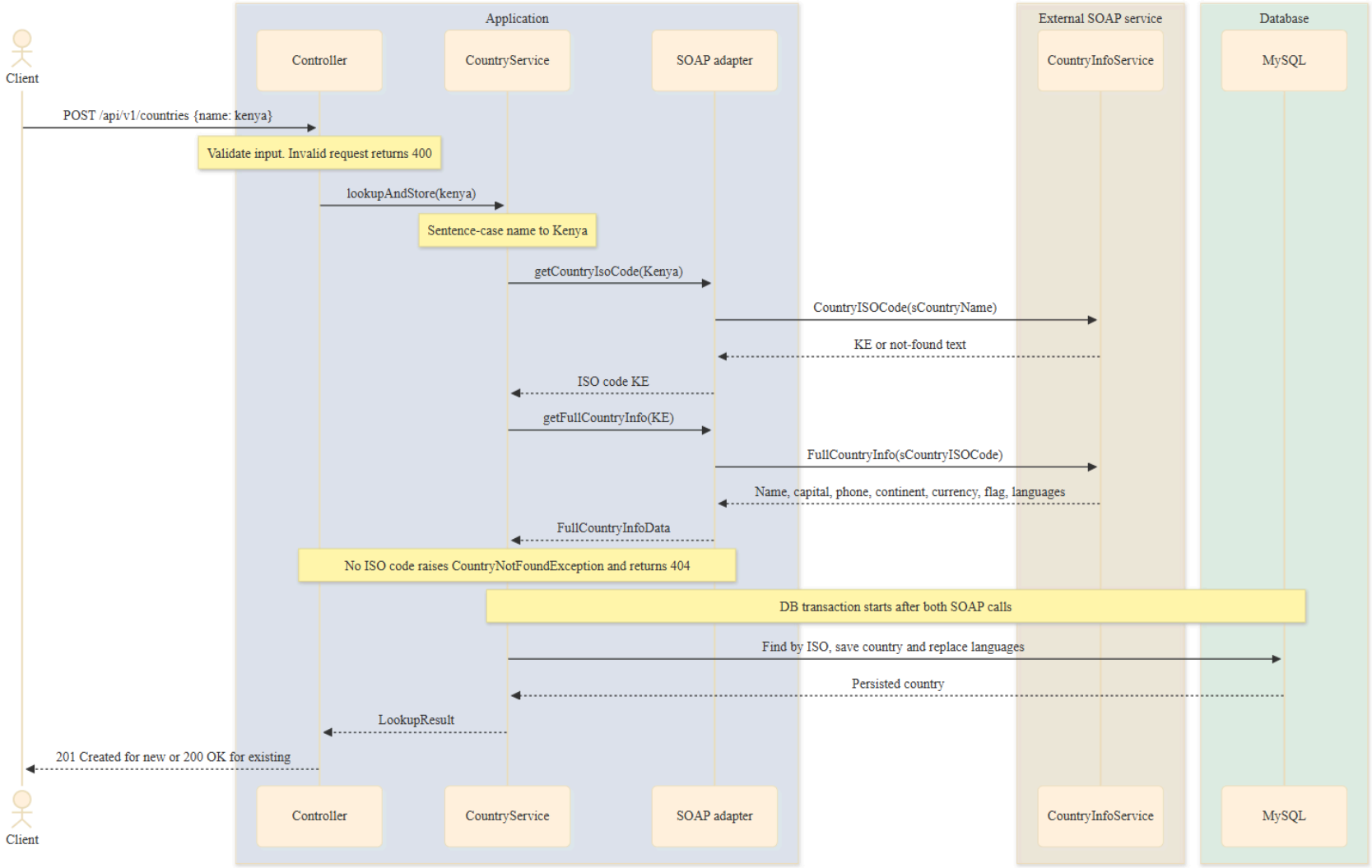
https://github.com/harryshirlif21/ncba-kubernates

Kubernetes deployment architecture



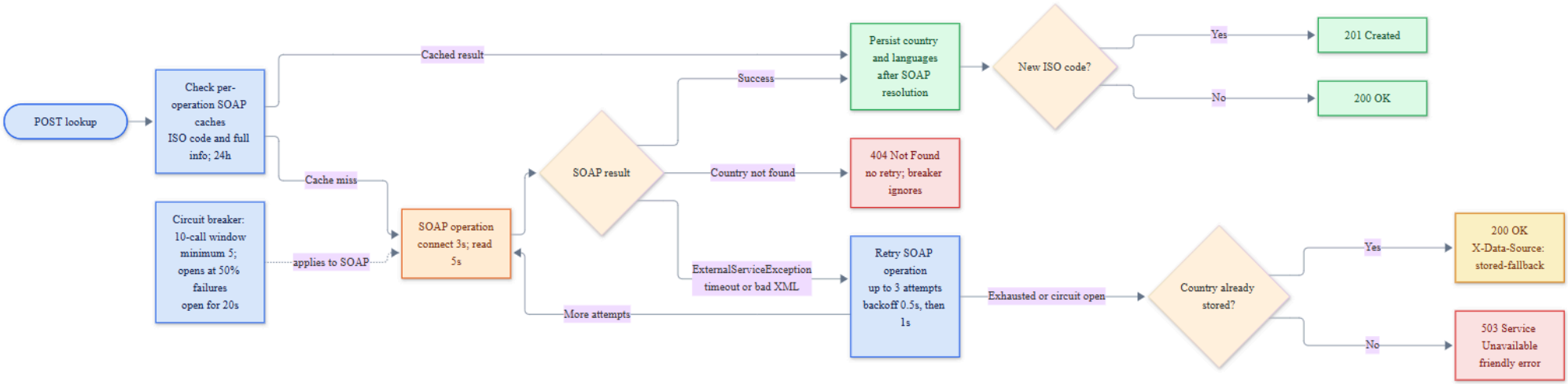
Shows ingress and service routing, the application Deployment and policies, MySQL persistence, and the external SOAP boundary.

Request flow: POST /api/v1/countries



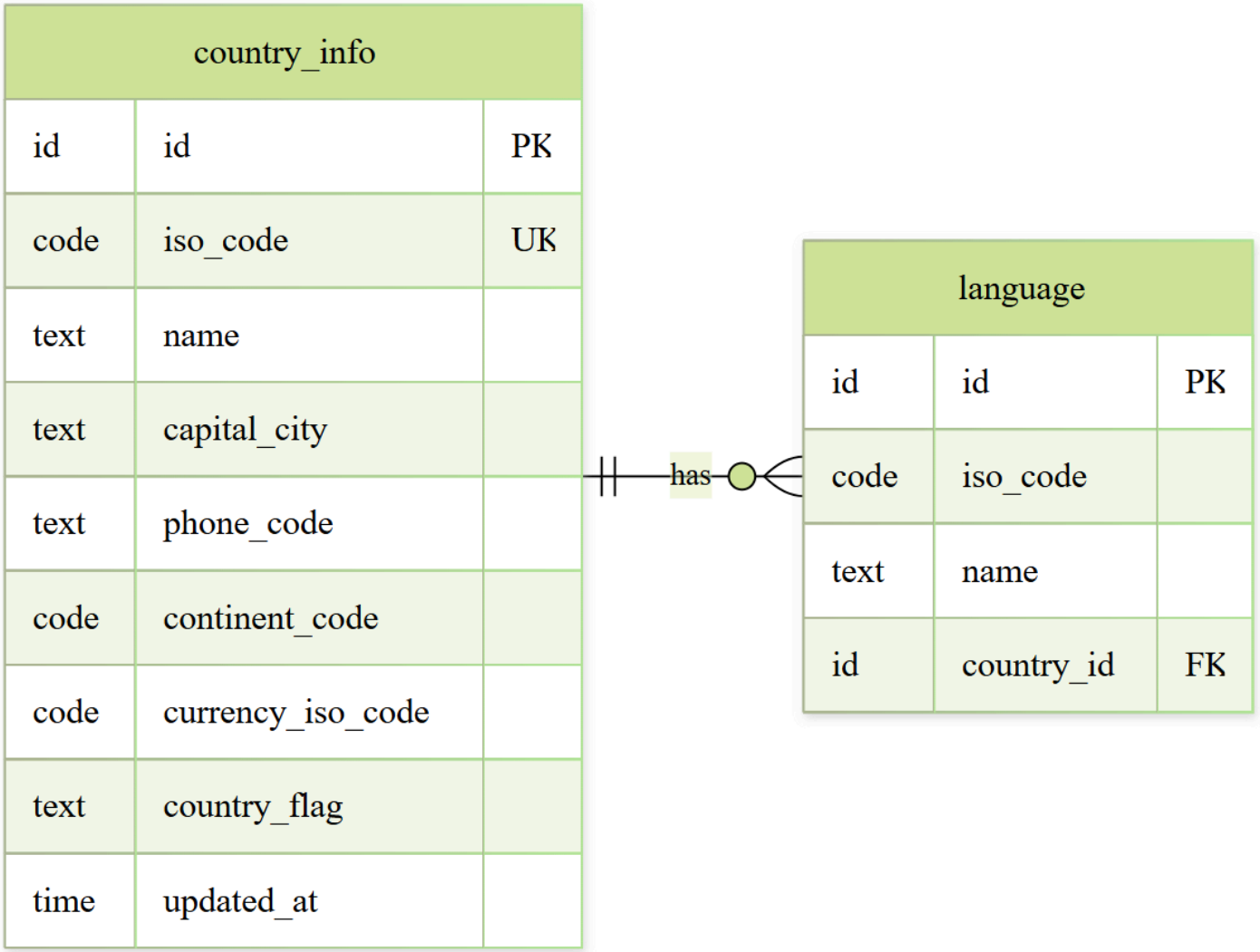
Follows validation, the two sequential SOAP calls, persistence after SOAP resolution, and the 201 or 200 response.

Resilience and failure handling



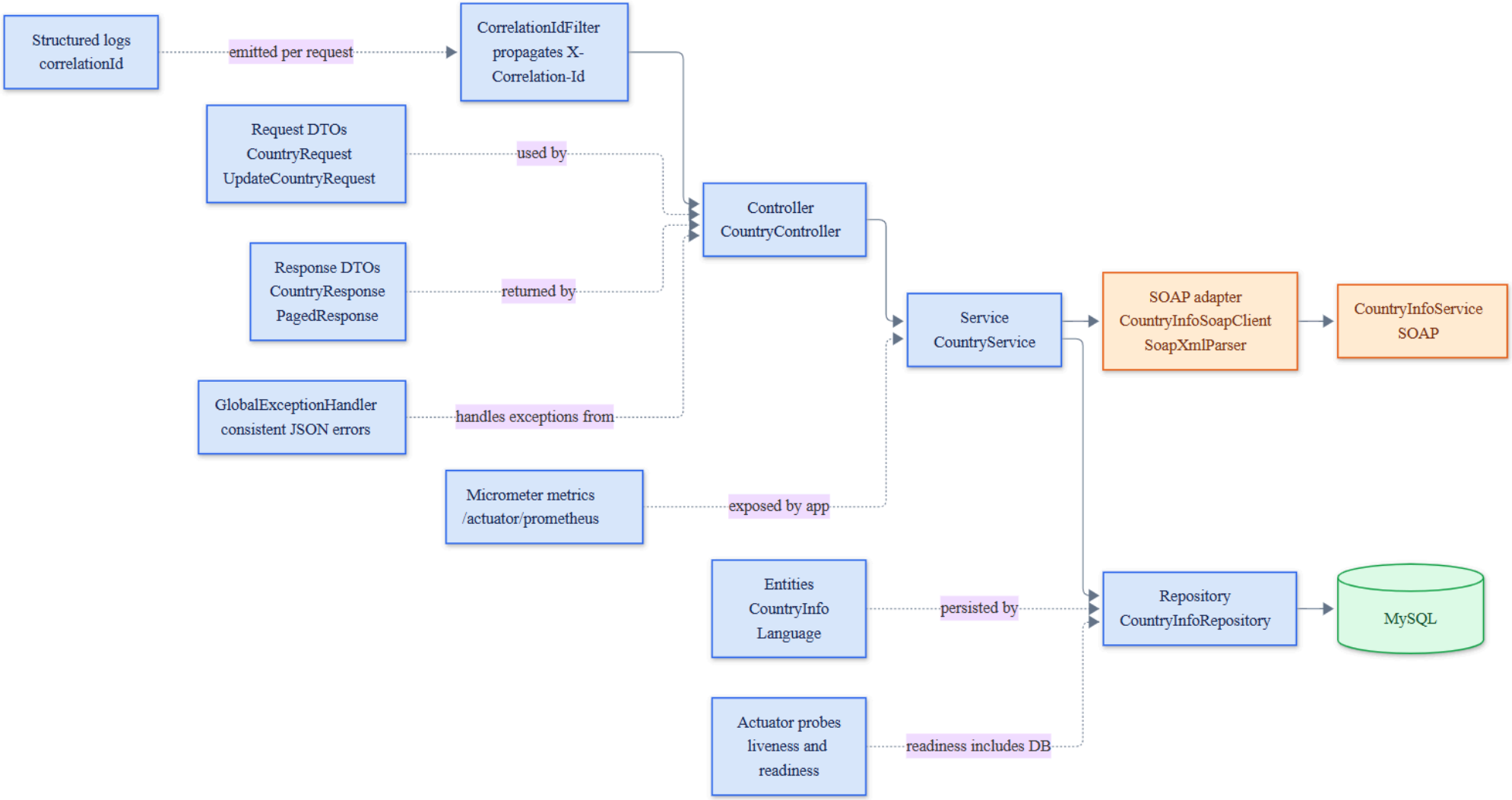
Shows SOAP caching, timeouts, retries and circuit breaking, plus the stored-country fallback and 404 or 503 outcomes.

Data model



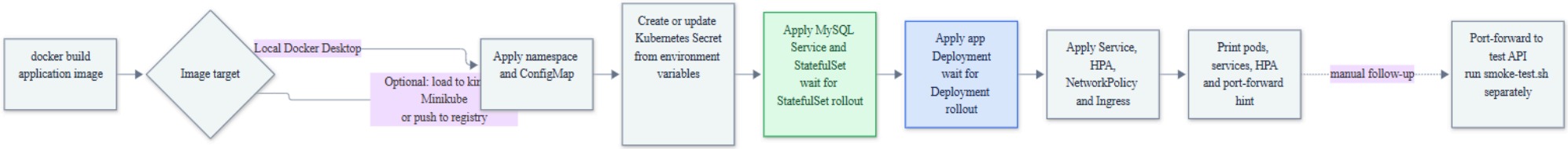
Shows the one-to-many country-to-language relationship, its foreign key, and cascading language deletion.

Application layers (MVC)



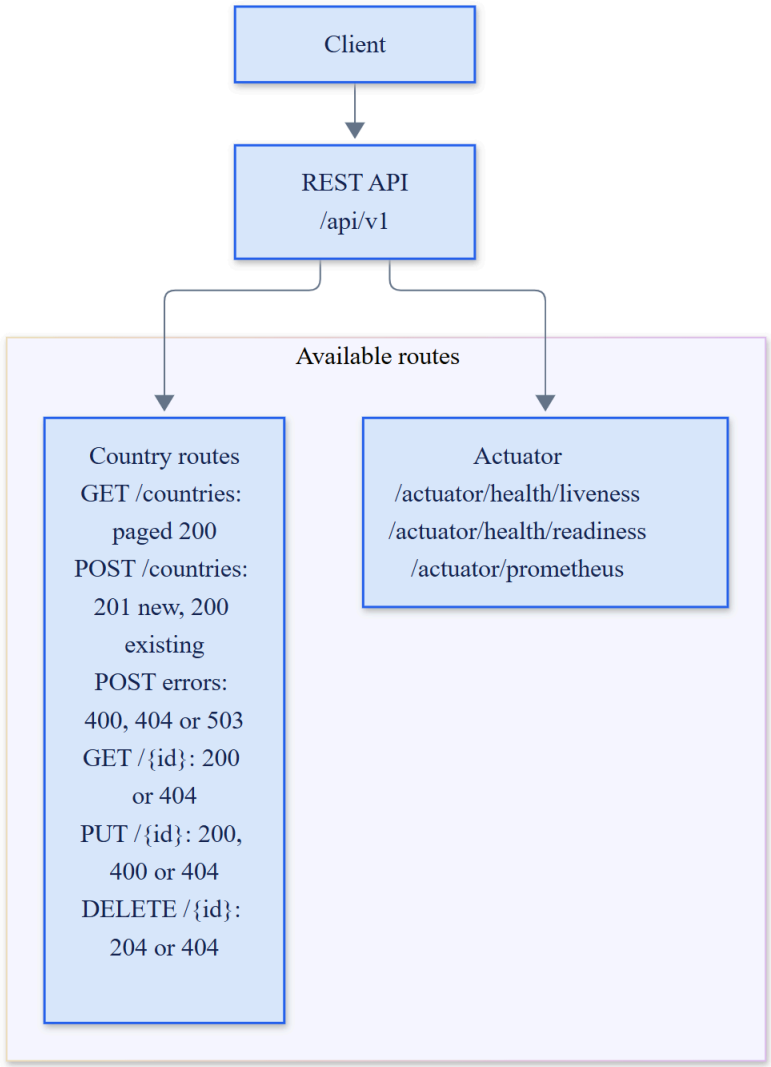
Maps the request filter, controller, service, repository and SOAP adapter to their DTOs, entities and cross-cutting services.

Deployment pipeline (scripts/deploy.sh)



Shows the script’s image and Kubernetes resource rollout. Port-forwarding and smoke testing are separate manual follow-up steps.

REST API endpoints



Summarizes the country CRUD routes and the exposed liveness, readiness and Prometheus endpoints.